

Zeroth-order and memory-efficient optimization for large language model fine-tuning and analog/RF circuit design.

## EDUCATION

|                                                                                                        |                             |
|--------------------------------------------------------------------------------------------------------|-----------------------------|
| University of California, Santa Barbara, <i>Ph.D. in Computer Engineering</i>   CA, USA                | Spring 2025 - Expected 2028 |
| University of California, Santa Barbara, <i>M.S. in Computer Engineering</i>   CA, USA                 | Fall 2023 - Winter 2025     |
| University of California, Santa Barbara, <i>Extension Program in ECE</i>   CA, USA                     | Fall 2022 - Spring 2023     |
| Huazhong University of Science & Technology, <i>B.S. in Electronic Information Engineering</i>   China | Fall 2019 - Summer 2023     |

## INDUSTRIAL EXPERIENCE

Cadence Design Systems, *Software Architect Intern* | Austin, TX Jun 2025 - Sep 2025

- Architected an LLM copilot agent for Voltus, Cadence's power-integrity signoff solver, turning natural-language design intent into verified command invocations, GUI actions, and automated root-cause analysis over signoff reports.
- Enabled on-premise deployment under enterprise data constraints through retrieval over EDA documentation, parameter-efficient fine-tuning, and distillation of a large teacher into a compact local model.

## SELECTED PUBLICATIONS

**GRZO: Group-Relative Zeroth-Order Optimization for Large Language Model Fine-Tuning** [paper]

Liyan Tan, Yequan Zhao, Yifan Yang, Ruijie Zhang, Xinling Yu, Zheng Zhang, *Findings of the Association for Computational Linguistics: EMNLP*, 2026.

**ZOAF: Towards Efficient Zeroth-Order Optimization for Analog/RF Circuit Design** [paper]

Liyan Tan, Yequan Zhao, Jinming Lu, Ben F. Jamroz, Ari Feldman, Zheng Zhang, under review at *IEEE TCAD*, arXiv:2606.02869, 2026.

**IAPO: Input Attribution-Aware Policy Optimization for Tool Use in Small Multimodal Agents** [paper]

Yifan Yang, Zhen Zhang, Jiayi Tian, Liyan Tan, Zheng Zhang, *arXiv preprint arXiv:2606.11652*, 2026.

**FuRA: Full-Rank Parameter-Efficient Fine-Tuning with Spectral Preconditioning** [paper]

Yequan Zhao, Ruijie Zhang, Liyan Tan, Niall Moran, Tong Qin, Zheng Zhang, *arXiv preprint arXiv:2605.22869*, 2026.

**MUON+: Towards More Effective Muon via One Additional Normalization Step for LLM Pre-training** [paper]

Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Yupeng Su, Liyan Tan, Zheng Zhang, *arXiv preprint arXiv:2602.21545*, 2026.

**Overview and Idea of the Mars UAV Experimental Platform Development** [paper]

Liyan Tan, Yijun Mao, *Advances in Aeronautical Science and Engineering*, 15(2):1-10, 2024.

## RESEARCH PROJECTS

**Variance-Reduced Zeroth-Order Optimization for LLM Fine-Tuning** | Advisor: Zheng Zhang Mar 2025 - Present

- Proposed GRZO, a group-relative zeroth-order optimizer that assigns a pseudo-independent perturbation to every mini-batch example, raising the effective gradient-direction count from one to the batch size at no additional forward cost.
- Proved directional unbiasedness with variance shrinking proportionally to the batch size; at the same inference-level memory footprint and comparable wall-clock cost, improved Llama3-8B average accuracy by +3.0 over MeZO and lifted sparse, low-rank, and quantized ZO variants by +4.9 on average.

**Zeroth-Order Optimization for Analog/RF Circuit Design** | Advisor: Zheng Zhang, with NIST Jan 2025 - Present

- Proposed ZOAF, a surrogate-free framework recovering gradient-descent directions from a small number of black-box circuit simulations, with a hybrid schedule from random-direction exploration to coordinate-wise refinement, quasi-random multi-start, and a sliding-window monitor for early stopping and feasibility.
- Achieved the best median figure of merit on all three benchmark schematics, with up to an order-of-magnitude advantage in median peaking on a 22-parameter amplifier and 1.3-3.8 $\times$  fewer simulator calls to convergence.
- Extending the framework toward uncertainty-aware design: stochastic zeroth-order optimization to handle process variations in nanoscale circuits, and distributionally robust circuit optimization that accounts for distribution shift in those variations.

**Tensor-Train Compressed NN via Temporal Logic** | Advisors: Zheng Zhang, Tim Sherwood Apr 2024 - Jan 2025

- Implemented tensor-train compressed neural networks on temporal-logic hardware using delay-space values and approximated arithmetic, reducing parameter count by  $115\times$  at 2.21% (exact) and 4.72% (approximated) accuracy loss.
- Cut latency by  $14.28\times$  and energy by  $32.37\times$  on a  $4 \times 4$  PE array under race-logic hardware settings.

**Mars UAV Aerodynamic Performance Experimental Platform** | Advisor: Yijun Mao, HUST Jan 2021 - Sep 2022

- Surveyed two decades of Mars UAV experimental platforms and summarized their design trade-offs; the resulting review was published in *Advances in Aeronautical Science and Engineering*.
- Built a hover-efficiency test platform with an axially distributed model, measuring rotational speed, torque, lift, and induced velocity of the rotor blades.

## TEACHING

---

**Teaching Assistant, UC Santa Barbara:** PHYS 127BL Digital Electronics; PHYS 127AL Analog Electronics; PHYS 6BL Introductory Physics; ECE 15A Fundamentals of Logic Design; PSTAT 5A Understanding Data.

## HONORS

---

**Honours Degree for Undergraduates,** Huazhong University of Science and Technology

Jun 2023